NEURAL ENGINEERING LAB (86-783 / 42-783 / 24-777)
EEG OSCILLATIONS LAB B : EOG AND ALPHA WAVES

Date: 9/19/23
Module: EEG

1) Objectives

Place electrodes to capture EEG and troubleshoot sources of noise during EEG signal acquisition
Filter, denoise and smooth EEG signals
Isolate the prominent features in EEG signals to verify quality of data
Improve EEG signal acquisition, feature observation and filtering
Track eye movements through a use-case of EEG specifically called Electro-oculography (EOG)
Manipulate alpha power (observed in Lab A) with different experimental conditions

2) Advice

Please read through the instructions in this document thoroughly and complete each step before moving on.
You are encouraged to discuss ideas and questions with the members of the teaching team during the lab.
If you get stuck at any point, first debug it yourself using the Setup & Troubleshooting guide and then reach out to members of the teaching team.
Identify one member of your group to serve as the group leader for each report. The group leader will be responsible for uploading the report to Canvas.

Please ask for help if you are unsure about any of the questions!

3) Grading

Please refer to the lab report guidelines on Canvas while preparing your reports.

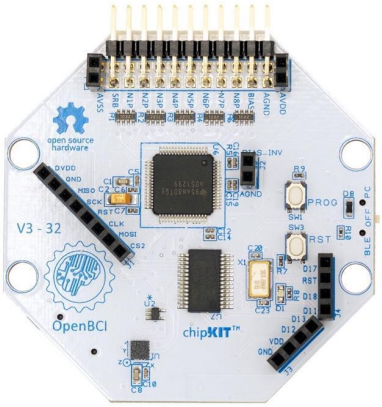
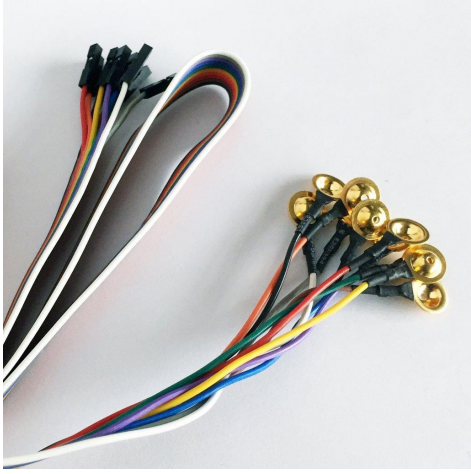
Each group will submit a single report as a PDF document using the following naming convention for your report: *GroupX_EEGOscillationsLab.pdf*, where *X* = your group number

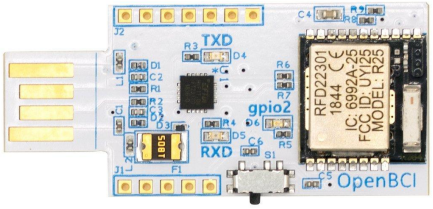
Submit your group report via the Assignments tab in Canvas.

The cover page for your report should list the following:

> **Title of the lab: EEG Oscillations Lab B - EOG and alpha waves**
>
> **Date of the lab:**
>
> **Group number:**
>
> **Names of group members and contribution statements.** The contribution statement should be written by each member describing their contribution to the report (e.g. Jane was primarily responsible for writing the analysis code and also contributed to data analysis and narrative responses for questions 2 and 6).

4) Components and supplies

OpenBCI Cyton Board and personal laptop (with power adapter)	
Gold cup electrodes	
Ten20 Conductive Paste	

Cyton Dongle (and a free USB port on your computer)	
First aid paper tape	

5) Theory

In the first part of this laboratory sequence, you read in EEG signals from two channels with the OpenBCI system, and saw the impact of eye movements and blinks on frontal EEG, as well as the magnitude of alpha in occipital EEG.

The front of the eye (cornea) and back of the eye (retina) form a dipole that creates a field potential, sometimes called the corneo-fundal potential. As the eye moves, that field potential's effect on an electrode can change (remember physiological artifacts observed in Lab A). This is the basis for a type of recording known as an EOG, or electrooculogram[1]. The cornea is positive relative to the retina, so as the eye moves toward your recording electrode, the potential should shift positive.You can do an EOG recording with as few as two electrodes, one on each side of the eye. These could be left/right, or up/down, depending on what type of eye movements you want to capture.

In the second part of the lab, you're going to do one of the classic Adrians experiments we talked about in class. [2]

References

[1]Jia and Tyler (2019) Measurement of saccadic eye movements by electrooculography for simultaneous EEG recording. Behav Res Methods. 2019; 51(5): 2139–2151. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6797659/

[2]Bazanova and Vernon (2014) Interpreting EEG alpha activity. Neuroscience & Biobehavioral Reviews Volume 44, July 2014, Pages 94-110. https://doi.org/10.1016/j.neubiorev.2013.05.007

[3]Belkhiria and Peysakhovich (2020) Electro-Encephalography and Electro-Oculography in Aeronautics: A Review Over the Last Decade (2010–2020). Front. Neuroergonomics, 21 December 2020. https://www.frontiersin.org/articles/10.3389/fnrgo.2020.606719/full

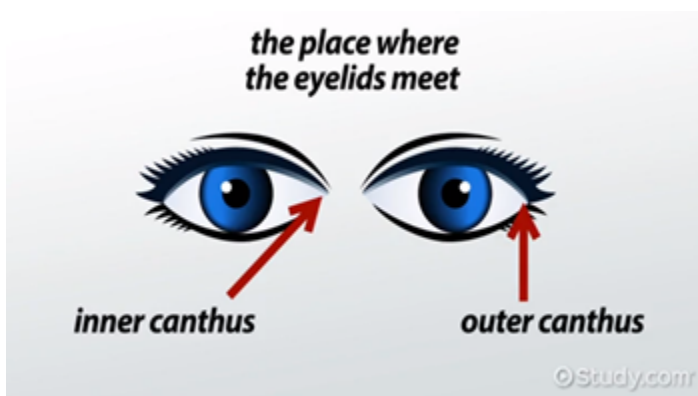
6) Procedures

⚠Connect OpenBCI hardware to your computer (Refer to Lab 0 & Troubleshooting Guide). Ensure proper connection and test the accelerometer signal and close the OpenBCI GUI.⚠

A) Eye Tracking with EOG

1) We will be focusing on the horizontal movements of the eyes, so the connections (to be made on one eye) will be:
   1st electrode – outer canthus → channel 1
   2nd electrode – inner canthus → SRB1
   3rd electrode – forehead just above the eye → BIAS
   (Feel free to use any color of wire, but ensure that you remember which is which. As a self-rule, I usually keep the SRB and BIAS wires as white and black respectively to avoid confusion.)

(You should be able to record eye movement and blink signals.)

Figure 1: Eyes and canthus	

2) Finally, load up the OpenBCI GUI and start a session. *(You can disable the channels you're not using (which helps with visualization) by clicking the number button for each of them to the left of their trace.* **Additionally, you may also try shorting the unused channels by clicking on "Hardware Settings", and changing the Input type from Normal to Shorted; this should further reduce crosstalk and give cleaner signals.** *)*
3) Now, set up 7 targets in front of the subject, at 10 degrees of visual angle spacing (the width of your fist at arm's length is about 10 degrees). The main aim is to have the targets spaced along somewhere evenly to cover the visual field. .

4) Recording: Record the subject looking at each target and holding their eye position there for about a second (you may want to use a metronome or other timer for the subject in this task). Perform about 5 trials of looking through each of the 7 targets in sequence, recording this in the data file.

For your lab report:
1 – read in this data and filter it
2 – align the segments of the voltage trace, and make a plot in which you show the relationship between voltage on the EOG (on the y-axis) and eye position (on the x-axis)
3 – fit a line to this, and determine the relationship between voltage and degrees of visual angle.
4 – with this line, you could get calibrated eye position data from the EOG. Plot this relationship with the data in your lab report.

⚠ Note: For filtering in this case, you may want to use a lowpass filter. You can also try bandpass with a very low bottom value (perhaps 0.01 or 0.001). The normal 1-50Hz range may not fully capture the show shifts in DC potential on the array that is seen in EOG. Either way, whatever filter you use you want to preserve the "staircase" look to the raw voltage of the EOG signal. ⚠

B) Alpha oscillations

1) Return to the configuration as in Lab A:
   White – right earlobe → SRB
   Black – left earlobe → BIAS
   Purple – forehead → N1P
   Red – back of your head about an inch to the side of center → N2P
   You could even do more than one occipital electrode if you want (but that's up to you).
   (Setup this configuration as you did before, and make sure you can observe alpha when the subject's eyes are closed.)
2) Recording: Measure alpha oscillations in at least three experimental conditions.
   a) With eyes open, resting
   b) With the eyes closed, resting
   c) With the eyes closed, but concentrating **(Be creative in how you define the conditions. Is there a particular way you want to ensure the subject is concentrating?)**
3) For each of these conditions, you will take 20 recordings of data (each a duration of 15 seconds), and keep track of the conditions for each of your recordings. *(This is to ensure that we have sufficient good data to perform analysis.)*
4) Once you have recorded these files, your goal will be to analyze the power spectrum of each condition, isolate the alpha power, and determine whether you find evidence that concentration has an effect on alpha oscillation power.
5) You'll need to calculate alpha power in a range (let's say 8-12 Hz) for each time series. You could simply bandpass filter the signal in the 8-12 Hz range and then square it. There are other ways to do this that you can choose. But in the end, what you need to do is show your work either mathematically or in code.

For your lab report:
1 – plot the raw data is at least one condition
2 – show how you calculated alpha power for that data
3 – divide the data for each condition into epochs such that you can calculate alpha power in each epoch
4 – plot the average alpha power across epochs, with error bars showing the standard error of the mean, for each condition
5 – use a statistical test to ask whether any differences between the conditions are statistically significant

7) Result

Great work! You have achieved Level 2 Mastery in EEG by observing eye movements in your EEG and finding whether or not there is a correlation between concentration and alpha oscillations. Don't forget the lab reports!

Please refer to the lab report guidelines on Canvas while preparing your reports.